

Complete Record Edges, Path Holonomy, and Recovery-Correctable Readouts

Tau Core Foundation Paper II

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Abstract

We isolate a finite technical question arising in morphological readout frameworks: under what conditions can a source-side action separation be identified exactly with an observer-carrier Uhlmann chord? A reversible *complete record edge* is defined as a source-owned, trace- and dagger-preserving relation between occupied finite record algebras, equipped with one common action unit and no terminal bypass. Primitive-defect extensivity then forces an exact Trace–Gram law on the supported endpoint pair. Composable complete edges form a groupoid, but endpoint-only descent is valid only when relative holonomy is null under a source-frozen operational descriptor or belongs to a declared Uhlmann gauge.

We then separate this reversible theorem from irreversible record formation. A three-qubit bit-flip construction gives an explicit non-isomorphic CPTP map with an exact recovery on its occupied logical operator system. Monotonicity in both directions proves preservation of fidelity and hence of the Uhlmann chord on that system. By contrast, commuting records alone cannot distinguish reversible transport from dephasing. This yields a seven-part occupation certificate that adds either a coherent off-diagonal witness or a certified recovery test to independent action, tomography, and timing measurements. For nonexact recovery we derive an explicit Uhlmann-chord loss bound from a uniform trace-norm error, and hence from a full-code diamond-norm certificate. The results are representation and identifiability theorems. They do not prove that a physical parent source creates or occupies complete edges, that a laboratory carrier realizes the proposed common-action law, or that Tau Core is empirically validated.

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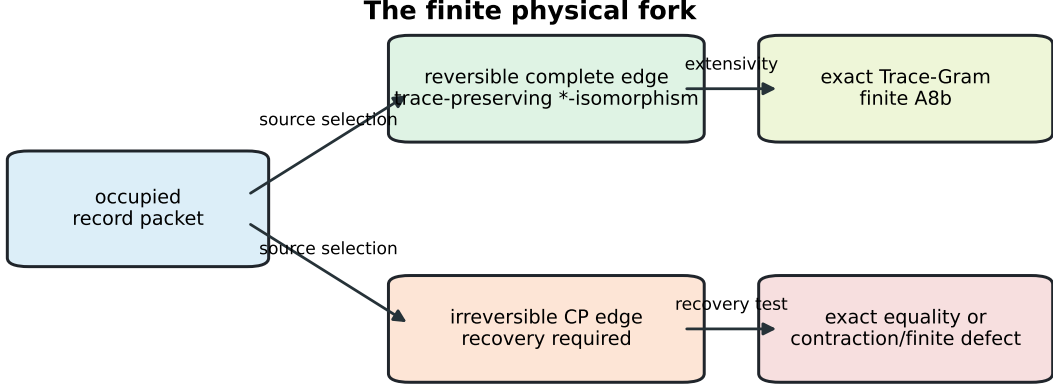


Figure 1: The finite physical fork. Exact Trace–Gram transport follows on a reversible complete edge. An irreversible CP relation requires a recovery theorem; otherwise contraction or an additional finite defect remains.

1 Question and claim boundary

Let a stabilized source/body construction assign a dimensionless finite action separation

$$x_A = \frac{\mathcal{X}_s}{\mathcal{A}_*}, \quad (1)$$

while an observer carrier assigns density operators ρ_u, ρ_v and the minimum-lift Uhlmann chord

$$x_Q = d_{U, \text{chord}}^2(\rho_u, \rho_v) = \min_U \|W_v - W_u U\|_{\text{HS}}^2 = 2(1 - \sqrt{F(\rho_u, \rho_v)}). \quad (2)$$

The target identity is $x_A = x_Q$. Equality of infinitesimal Hessians is not enough: a finite response can have the same tangent metric but differ by nonlinear terms. This paper therefore asks which *finite source relations* are sufficient for the identity, and how this statement changes for irreversible completely positive (CP) maps.

The status is deliberately narrow:

- complete-edge existence is an assumption about an occupied source relation, not a theorem about Nature;
- the Trace–Gram result is exact only on the declared occupied support;
- descriptor-null holonomy is operational and claim-relative, not an ontological erasure rule;
- the CP example is a recovery-correctable toy model, not an identified physical Tau carrier;
- no numerical constant, Standard-Model spectrum, or empirical anomaly is derived here.

2 Finite record packet

Let the occupied record support decompose into finitely many central blocks,

$$\mathcal{A}_Q = \bigoplus_{\alpha=1}^m M_{n_\alpha}(\mathbb{C}), \quad \tau(a) = \sum_{\alpha} p_{\alpha} \frac{\text{Tr}(a_{\alpha})}{n_{\alpha}}, \quad p_{\alpha} > 0, \quad \sum_{\alpha} p_{\alpha} = 1. \quad (3)$$

We assume a finite special dagger-Frobenius packet $(\mathcal{A}_Q, \mu, \eta, \Delta = \mu^\dagger, \epsilon = \eta^\dagger)$ with

$$\mu\Delta = I_{\mathcal{A}_Q}. \quad (4)$$

Specialness makes the primitive copy map isometric in the inherited trace metric. It is a strong normalization condition; a merely normalizable Frobenius algebra has $\mu\Delta = L$ for a positive central loop operator L , leaving a possible block-dependent scale.

Definition 1 (Finite recovery completeness). *A source law is finite recovery-complete on an occupied family when its finite action separation depends only on the source-owned minimum-lift chord,*

$$x_A = F(x_Q), \quad F(0) = 0, \quad (5)$$

and no additional positive stiffness remains on the declared source support while being invisible to that lift. This is a sufficient-statistic assumption for finite action separation, not a consequence of a positive local Hessian alone.

The admissible domain for exact extensivity is

$$\mathcal{D}_2 = \{(Y_1, Y_2) : Y_i \geq 0, Y_1 + Y_2 \leq 2, Y_1, Y_2 \text{ arise from admitted orthogonal source slots}\}. \quad (6)$$

The upper bound follows from the squared chord convention. Extensivity means

$$F(Y_1 + Y_2) = F(Y_1) + F(Y_2), \quad (Y_1, Y_2) \in \mathcal{D}_2, \quad (7)$$

with one universal continuous F on the connected occupied orbit. Merely assigning separate functions F_α to blocks would not imply a common finite law.

Lemma 1 (Trace-Gram response on the admissible domain). *If Eq. (5) is continuous, has local unit $F'(0) = 1$, and satisfies Eq. (7) on a connected admissible interval, then $F(Y) = Y$ throughout that interval.*

Proof. The restricted Cauchy equation gives rational homogeneity on every admissible subinterval. Continuity extends it to real arguments. Hence $F(Y) = cY$, and $F'(0) = 1$ fixes $c = 1$. $\square$

The assumptions are material. The family

$$F_\lambda(Y) = Y + \lambda Y^2 \quad (8)$$

has the same local unit but composition defect

$$F_\lambda(Y_1 + Y_2) - F_\lambda(Y_1) - F_\lambda(Y_2) = 2\lambda Y_1 Y_2. \quad (9)$$

Thus local metric agreement does not establish finite A8b.

3 Complete source-owned record edges

Definition 2 (Complete record edge). *Let u, v be occupied fibers with finite special dagger-Frobenius record algebras $(\mathcal{A}_{Q,u}, \tau_u)$ and $(\mathcal{A}_{Q,v}, \tau_v)$. A source relation $e : u \rightarrow v$ is a complete record edge on a declared support when its realization:*

(E1) is an invertible multiplication-, unit- and dagger-intertwiner;

(E2) preserves the inherited metric, faithful trace and common action unit;

(E3) transports the full declared endpoint record pair;

(E4) is source-owned and contains no terminal-only bypass copy; and

(E5) belongs to the same primitive-extensive occupied edge orbit.

This definition characterizes sufficient structure. It does not derive edge existence or invertibility.

Proposition 1 (Exact finite identity on a complete edge). *On a finite recovery-complete, primitive-extensive packet, every occupied complete edge restricts to a faithful trace-preserving $*$ -isomorphism*

$$\Pi_e : \mathcal{A}_{Q,u} \rightarrow \mathcal{A}_{Q,v}, \quad \tau_v(\Pi_e(a)) = \tau_u(a), \quad (10)$$

whose standard-form lift is isometric. For its supported endpoint pair,

$$\boxed{\frac{\mathcal{X}_s(e)}{\mathcal{A}_*} = \min_U \|W_v - W_u U\|_{\text{HS}}^2 = d_{U,\text{chord}}^2(\rho_u, \rho_v).} \quad (11)$$

Proof. The algebraic intertwiners and invertibility give a faithful $*$ -isomorphism. Metric and common-unit transport exclude a rescaled trace. Therefore

$$\langle L^2(\Pi_e)a, L^2(\Pi_e)b \rangle_v = \tau_v\left(\Pi_e(a)^\dagger \Pi_e(b)\right) \quad (12)$$

$$= \tau_v\left(\Pi_e(a^\dagger b)\right) = \tau_u(a^\dagger b), \quad (13)$$

so the standard-form lift is isometric. Source ownership places the amplitude residual in the once-counted primitive defect ledger. Finite recovery completeness and the previous lemma then force $x_A = x_Q$. $\square$

A rescaled trace, a terminal-only amplitude copy, or the nonlinear response (8) violates a distinct hypothesis. The theorem therefore packages a strong but auditable sufficient realization rather than showing that every stable record process has this form.

4 Operational descriptor completeness

Holonomy cannot be declared null by inspecting density matrices alone. The descriptor must be fixed before paths are compared.

Definition 3 (Claim-complete operational descriptor). *For an occupied fiber u , a descriptor protocol $\mathfrak{D}_u^{(C)}$ is complete relative to claim C when it freezes:*

(D1) the occupied operator system $\mathcal{S}_{\text{occ}}$;

(D2) an informationally complete effect frame on that system, or an explicit restricted-observable label;

(D3) estimators for the retained action and metric data;

(D4) every typed terminal and calibration used by C ;

(D5) the admitted gauge subgroup and path family;

(D6) equivalence margins δ_j , confidence level and multiple-test correction.

For two path outputs x_p, x_q , descriptor-nullity is accepted only by a predeclared equivalence test,

$$\text{CI}_{1-\alpha}[D_j(x_p) - D_j(x_q)] \subseteq [-\delta_j, \delta_j] \quad \text{for every frozen component } j. \quad (14)$$

Failure to reject equality is not evidence of nullity. Completeness here is claim-relative and never means access to all parent information.

If the effect frame is informationally complete on $\mathcal{S}_{\text{occ}}$, equality of all ideal probabilities identifies the state on that system. Finite data replace equality by Eq. (14). A central automorphism is not automatically gauge: block permutations, weights and typed roles remain physical whenever the frozen descriptor detects them.

Three-vertex descriptor-frozen holonomy control

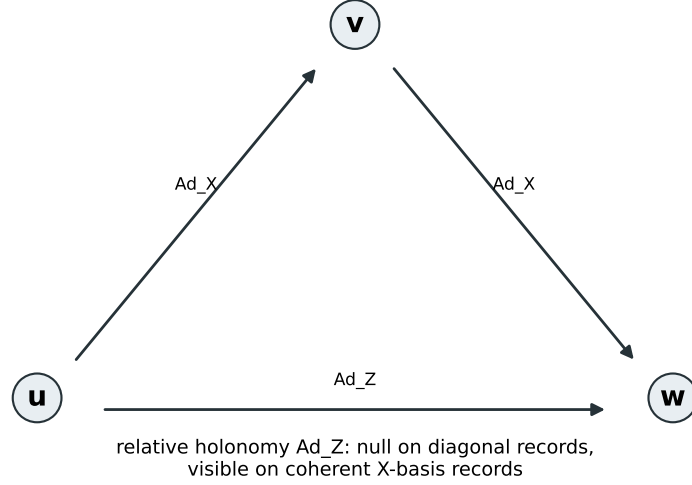


Figure 2: A three-vertex control. The same relative automorphism is null on a diagonal record family but visible once a coherent X -basis witness is frozen into the endpoint descriptor.

5 Paths and holonomy

Composable complete edges form a groupoid $\mathcal{G}_{\text{rec}}$. A path $p = e_n \circ \dots \circ e_1 : u \rightarrow v$ induces $\Pi_p = \Pi_{e_n} \circ \dots \circ \Pi_{e_1}$. For two paths $p, q : u \rightarrow v$, define relative holonomy

$$H_{p,q} = \Pi_q^{-1} \Pi_p. \quad (15)$$

Corollary 1 (Pathwise identity and endpoint descent). *Every complete path obeys Eq. (11) pathwise. An endpoint-only law descends on a declared occupied family exactly when $H_{p,q}$ belongs to the frozen Uhlmann gauge or is null under the claim-complete descriptor. Otherwise the path class is retained physical data.*

Proof. Faithful trace-preserving $*$ -isomorphisms are closed under composition and inversion. Right-unitary Uhlmann gauge leaves $WW^\dagger$ and the minimized chord invariant. Any descriptor-visible relative automorphism distinguishes the two transported records and cannot consistently be quotiented. $\square$

For a concrete control, take three fibers carrying $M_2(\mathbb{C})$, with normalized trace, and edges

$$\Pi_e = \text{Ad}_X : u \rightarrow v, \quad \Pi_f = \text{Ad}_X : v \rightarrow w, \quad \Pi_g = \text{Ad}_Z : u \rightarrow w. \quad (16)$$

For $p = f \circ e$ and $q = g$, $\Pi_p = \text{id}$ while $H_{p,q} = \text{Ad}_Z$. On diagonal densities and effects this holonomy is null. If $|+\rangle\langle+|$ and an X -basis effect are frozen into the descriptor, Ad_Z maps $|+\rangle$ to $|-\rangle$ and becomes visible. The null decision therefore belongs to the predeclared operational claim, not to an analyst's post-hoc choice.

6 Irreversible CP relations

A generic detector, decohering evolution or rank-reducing instrument is not a complete edge. If $\mathcal{C}$ is CPTP, fidelity monotonicity gives

$$F(\mathcal{C}(\rho), \mathcal{C}(\sigma)) \geq F(\rho, \sigma), \quad d_U^2(\mathcal{C}(\rho), \mathcal{C}(\sigma)) \leq d_U^2(\rho, \sigma). \quad (17)$$

Exact finite A8b can therefore be lost. It is restored on an occupied domain if the map is exactly recoverable there. The domain must be typed:

Definition 4 (Recovery domains). *Let $P\mathcal{H}$ be a code subspace. We distinguish:*

- (R1) *a state family $\mathfrak{S} \subseteq \mathcal{D}(P\mathcal{H})$;*
- (R2) *a unital self-adjoint linear operator system $\mathcal{S}_{\text{op}} \subseteq \mathcal{B}(P\mathcal{H})$;*
- (R3) *a correctable unital $*$ -subalgebra $\mathcal{A}_{\text{corr}} \subseteq \mathcal{B}(P\mathcal{H})$; and*
- (R4) *the full code algebra $\mathcal{B}(P\mathcal{H})$.*

State-family recovery means equality only on $\mathfrak{S}$. Operator-system recovery means equality of the linear maps on $\mathcal{S}_{\text{op}}$. Algebra recovery preserves all states and observables encoded by $\mathcal{A}_{\text{corr}}$. Full-code recovery is the special case $\mathcal{A}_{\text{corr}} = \mathcal{B}(P\mathcal{H})$. None of these scopes may be silently enlarged after the test.

Proposition 2 (Recovery sandwich). *Let $\mathcal{C}$ and $\mathcal{R}$ be CPTP maps satisfying*

$$(\mathcal{R} \circ \mathcal{C})(\rho) = \rho \quad (18)$$

for every state in a frozen family $\mathfrak{S}$. Then for every pair $\rho, \sigma \in \mathfrak{S}$,

$$F(\mathcal{C}(\rho), \mathcal{C}(\sigma)) = F(\rho, \sigma), \quad d_U^2(\mathcal{C}(\rho), \mathcal{C}(\sigma)) = d_U^2(\rho, \sigma). \quad (19)$$

Proof. Monotonicity under $\mathcal{C}$ gives one inequality. Applying monotonicity under $\mathcal{R}$ and using Eq. (18) gives the reverse inequality. Equality of fidelity implies equality of the chord convention in Eq. (2). $\square$

This statement is closely related to quantum error-correction conditions [1] and to the broader sufficiency/recovery viewpoint [2], but only the explicit recovery identity is needed below.

Proposition 3 (Explicit approximate-recovery chord bound). *Let $\mathfrak{S}$ be a frozen state family and suppose*

$$\sup_{\rho \in \mathfrak{S}} \|(\mathcal{R}\mathcal{C})(\rho) - \rho\|_1 \leq \eta. \quad (20)$$

This follows, in particular, from $\|\mathcal{R}\mathcal{C} - \text{id}\|_{\diamond} \leq \eta$ on the full code algebra. For $\rho, \sigma \in \mathfrak{S}$, define

$$x_{\text{in}} = d_U^2(\rho, \sigma), \quad x_{\text{out}} = d_U^2(\mathcal{C}\rho, \mathcal{C}\sigma). \quad (21)$$

Then

$$0 \leq x_{\text{in}} - x_{\text{out}} \leq x_{\text{in}} - [\max\{0, \sqrt{x_{\text{in}}} - 2\sqrt{\eta}\}]^2. \quad (22)$$

Equivalently,

$$x_{\text{in}} - x_{\text{out}} \leq \begin{cases} x_{\text{in}}, & x_{\text{in}} \leq 4\eta, \\ 4\sqrt{\eta x_{\text{in}}} - 4\eta, & x_{\text{in}} > 4\eta. \end{cases} \quad (23)$$

Proof. Write $d(\rho, \sigma) = \sqrt{d_U^2(\rho, \sigma)}$, the Bures chord metric. The Fuchs–van de Graaf inequality [3] gives

$$d(\rho, \mathcal{R}\mathcal{C}\rho)^2 = 2(1 - \sqrt{F(\rho, \mathcal{R}\mathcal{C}\rho)}) \leq \|\rho - \mathcal{R}\mathcal{C}\rho\|_1 \leq \eta, \quad (24)$$

and likewise for σ . The triangle inequality and contractivity under $\mathcal{R}$ imply

$$\sqrt{x_{\text{in}}} \leq \sqrt{x_{\text{out}}} + 2\sqrt{\eta}. \quad (25)$$

Contractivity under $\mathcal{C}$ also gives $x_{\text{out}} \leq x_{\text{in}}$. Solving these two inequalities for x_{out} yields Eqs. (22) and (23). $\square$

The bound is useful only relative to the frozen domain. It becomes weak when $x_{\text{in}} \lesssim 4\eta$, precisely where recovery error can hide the full input separation. It still concerns only the Uhlmann side; it does not make the once-counted source action equal to either chord.

The estimate is exact in the limit $\eta = 0$ and becomes vacuous in the first branch when $x_{\text{in}} \leq 4\eta$. It is a uniform worst-case consequence of trace-norm recovery plus metric contractivity; no claim of universal optimality is made for a specified channel, code, or state pair. A smaller state-dependent error may give a sharper bound, but it must be certified on the same predeclared state family. A diamond-norm certificate applies to the full code including reference systems, whereas the state-family hypothesis in Eq. (20) must not be silently promoted to that stronger scope.

7 A recovery-correctable CP toy model

Let $\mathcal{H}_L = \mathbb{C}^2$, $\mathcal{H}_P = (\mathbb{C}^2)^{\otimes 3}$, and define the encoding isometry

$$V|0\rangle = |000\rangle, \quad V|1\rangle = |111\rangle. \quad (26)$$

Let X_i flip physical qubit i , $P_0 = VV^\dagger$, and $P_i = X_i P_0 X_i$. These four syndrome subspaces are mutually orthogonal. For $0 \leq p \leq 1$, define

$$\mathcal{C}_p(\rho) = (1-p)V\rho V^\dagger + \frac{p}{3} \sum_{i=1}^3 X_i V\rho V^\dagger X_i. \quad (27)$$

This is CPTP and, for $0 < p < 1$, is not a $*$ -isomorphism from the logical matrix algebra onto its noisy image.

Define a decoder by syndrome measurement and correction,

$$\mathcal{R}_{\text{dec}}(\sigma) = V^\dagger \left[P_0 \sigma P_0 + \sum_{i=1}^3 X_i P_i \sigma P_i X_i \right] V + \text{Tr}(P_{\text{fail}} \sigma) |0\rangle\langle 0|, \quad (28)$$

where $P_{\text{fail}} = I - P_0 - \sum_i P_i$ makes the map trace preserving outside the occupied range.

Proposition 4 (Exact recovery-correctable non-isomorphic edge). *For every logical state ρ ,*

$$\mathcal{R}_{\text{dec}} \circ \mathcal{C}_p(\rho) = \rho. \quad (29)$$

Consequently Eq. (19) holds for all logical state pairs and all $p \in [0, 1]$.

Proof. Each summand of Eq. (27) lies in one orthogonal syndrome subspace. The corresponding term in Eq. (28) projects onto that subspace, applies the matching correction, and decodes with $V^\dagger$. Every branch therefore returns its probability weight times ρ ; the weights sum to one. The recovery-sandwich proposition gives fidelity and chord equality. $\square$

The toy model recovers the full logical algebra $\mathcal{B}(\mathcal{H}_L) = M_2(\mathbb{C})$, not merely a commuting state family. It proves that invertibility of the physical CP map is unnecessary for exact finite metric transport on a certified code. It does not show that the body action assigns the same x_A , nor that this code is a physical Tau carrier.

8 Why commuting records are insufficient

Let $\mathcal{D} \subset M_2(\mathbb{C})$ be the diagonal algebra and compare

$$\Pi_{\text{rev}} = \text{id}, \quad \Pi_{\text{irr}}(a) = P_0 a P_0 + P_1 a P_1. \quad (30)$$

Both maps agree on every diagonal density and diagonal effect. Yet Π_{irr} destroys off-diagonal coherence and has no CPTP inverse on the full matrix algebra.

Proposition 5 (Commuting-record reversibility no-go). *No protocol restricted to the commuting record algebra $\mathcal{D}$ can identify whether the occupied process is Π_{rev} or Π_{irr} . Reversibility requires either a coherent witness outside $\mathcal{D}$ or a certified recovery test on a larger predeclared operator system.*

Proof. The two channels induce identical probabilities for every state and effect in $\mathcal{D}$. Therefore every experiment confined to that algebra has the same likelihood under both hypotheses. An off-diagonal state/effect or a recovery test separates them. $\square$

9 FOC-7 occupation certificate

The preceding no-go upgrades the finite occupation certificate. A candidate carrier passes *FOC-7* only if one predeclared protocol establishes:

- (F1) a metastable, held, single-shot record with a frozen discrimination error;
- (F2) independent source-side reconstruction of x_A , without tomography or timing data;
- (F3) independent tomographic reconstruction of x_Q ;
- (F4) $x_A = x_Q$ within predeclared equivalence margins at at least two nonzero separations;
- (F5) either a coherent off-diagonal witness distinguishing the complete operator system from its commuting restriction, or a certified recovery map satisfying $\|\mathcal{RC} - \text{id}\|_\diamond \leq \eta_{\text{rec}}$ on that system;
- (F6) downstream timing with independently measured energy spread; and
- (F7) controls for decoherence, drift, detector response and nonlinear finite corrections.

Passing FOC-7 would identify the tested finite packet within its frozen model class. It would not by itself prove the entire parent ontology. Failure at F5 specifically prevents commuting data from being promoted into evidence for a reversible complete edge.

10 What remains open

The internal mathematics now distinguishes three cases:

$$\begin{aligned}
 \text{complete reversible edge} &\Rightarrow \text{exact finite Trace-Gram,} \\
 \text{recovery-correctable CP edge} &\Rightarrow \text{exact chord on the code,} \\
 \text{generic irreversible CP edge} &\Rightarrow \text{contraction or extra defect.}
 \end{aligned} \tag{31}$$

The decisive physical question is not settled by these implications:

Does the physical base-seed source create and occupy either the complete dagger-Frobenius edge packet or a recovery-correctable non-isomorphic packet, and does its once-counted action assign the same finite separation as the observer amplitude geometry?

The represented normalized trace is canonical after a representation is fixed, but its central weights are still physical input. Exact extensivity excludes interactions between admitted primitive defect slots and is likewise a strong source law. Finite recovery completeness says that the chord is a sufficient statistic for the action separation. These assumptions are now explicit and separately falsifiable; they are not smuggled in as consequences of local Hessian positivity.

Equation (22) controls approximate recovery on the frozen state family. It is not an exact A8b theorem: the finite source action remains independent, and experiment-specific promotion still requires a declared code, error model and reconstruction uncertainty.

11 Conclusion

Complete source-owned record edges provide a concise sufficient theorem for finite action–Uhlmann identity, while path holonomy states exactly when that identity can descend to endpoint-only data. The operational descriptor makes the null decision testable and prevents post-hoc removal of visible path information.

Irreversibility does not end the route: the three-qubit construction proves that a non-isomorphic CP map can preserve the Uhlmann chord on a certified recoverable code. But commuting record agreement is insufficient to decide between this structure, a reversible edge, and dephasing. FOC-7 therefore requires coherent or recovery evidence in addition to action, tomography and timing measurements.

The paper closes the internal finite alternatives, not their nature-level selection. Physical source occupation, common-action ownership and a laboratory carrier remain open empirical and source-theoretic problems.

A Fidelity convention

We use squared Uhlmann fidelity

$$F(\rho, \sigma) = \left[\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]^2. \quad (32)$$

The minimum Hilbert–Schmidt purification chord is

$$\min_U \|W_\rho - W_\sigma U\|_{\text{HS}}^2 = 2(1 - \sqrt{F}), \quad (33)$$

following Uhlmann’s purification geometry [4]. This is not the squared Bures distance convention used in every part of the literature; all equalities in this paper use the displayed definition.

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